

BIOLOGY 30

Unit 3A — Cellular Division Study Guide

1 Unit / Module Objectives

Topic Area	Diploma-Level Objective
Chromosome Number	Analyze chromosome number in somatic and sex cells, and justify the significance of haploidy (n), diploidy (2n), and polyploidy using centromere-based reasoning.
Cell Cycle	Explain the events of interphase, mitosis, and cytokinesis, and predict chromosome and chromatid number through each stage of the cell cycle.
Meiosis	Describe meiosis, including spermatogenesis and oogenesis, and justify why reduction of chromosome number is necessary before fertilization.
Mitosis vs. Meiosis	Compare mitosis and meiosis in terms of purpose, number of divisions, chromosome behaviour, genetic outcomes, and biological role.
Crossing Over and Nondisjunction	Evaluate how crossing over generates genetic variation and how nondisjunction alters chromosome number, gene dosage, inheritance, and development.
Twin Formation	Compare the formation of fraternal and identical offspring in a single birthing event and predict how genotype can remain constant while phenotype differs.
Reproductive Diversity	Interpret alternation of generations in organisms such as moss and pine by tracking dominant life stage, ploidy, and the location of meiosis.

2 Student Self-Verification Checklist

Topic	Self-Verification Statement
Chromosome Logic	☐ I can define chromosome number using centromeres and distinguish chromosome count from chromatid count.
Ploidy	☐ I can define haploid, diploid, and polyploid correctly and connect each term to biological examples.
Cell Cycle Tracking	☐ I can calculate chromosome number and chromatid number at G1, after S phase, metaphase, anaphase, and after cytokinesis.
Reduction Division	☐ I can justify why meiosis I is reductional and why meiosis II is equational.
Gametogenesis	☐ I can calculate the products of spermatogenesis and oogenesis and justify why cytokinesis is unequal in oogenesis.
Crossing Over	☐ I can interpret tetrads, chiasmata, and recombinant chromatids and justify why crossing over changes allele combinations but not chromosome number.

Nondisjunction	☐ I can calculate gamete outcomes for nondisjunction in meiosis I versus meiosis II and justify the pattern numerically.
Comparison Skills	☐ I can interpret the mechanistic differences between mitosis and meiosis and justify how those differences change the products.
Data Interpretation	☐ I can interpret cell-count or life-cycle data and calculate biologically valid conclusions from the evidence.
Twin Formation	☐ I can interpret how monozygotic and dizygotic twins form and justify why identical twins can still differ phenotypically.
Life Cycles	☐ I can interpret alternation of generations by identifying gametophyte and sporophyte stages and defining which stage is dominant.
Applied Biology	☐ I can justify how checkpoint failure or spindle disruption can lead to cancer-related cellular outcomes.

3 Diploma-Level Practice Questions with Fully Worked Solutions

Question	A species has a diploid chromosome number of $2n = 24$. (a) Determine the number of chromosomes and chromatids present in one cell at metaphase I of meiosis. (b) Determine the number of chromosomes and chromatids present in one gamete after meiosis II.
Step 1 — Identification	Identify the given diploid number and the exact stages named. Metaphase I occurs before homologous chromosomes separate. A gamete after meiosis II is haploid and no longer has duplicated chromosomes.
Step 2 — Principle	Chromosome number is counted by centromeres. Duplication changes DNA amount and chromatid number, but a duplicated chromosome still counts as one chromosome until sister chromatids separate.
Step 3 — Explicit Reasoning	Because $2n = 24$, the haploid number is $n = 12$. At metaphase I, the cell is still diploid and each chromosome is duplicated. Chromosomes = 24. Chromatids = $24 \times 2 = 48$. After meiosis II, each gamete is haploid. Chromosomes per gamete = 12. Because sister chromatids separated in meiosis II, each chromosome has one chromatid. Chromatids = 12.
Step 4 — Validation	Check the stage logic: reduction has not yet happened at metaphase I, so chromosome number remains 24. After meiosis II, the product must be haploid with single-chromatid chromosomes.
Step 5 — Final Answer	Metaphase I: 24 chromosomes and 48 chromatids. After meiosis II: 12 chromosomes and 12 chromatids.

Common Error	Confusing chromosome number with chromatid number. A duplicated chromosome has two chromatids but still counts as one chromosome.
Exam Strategy	Write n and $2n$ first, then ask whether the chromosomes are duplicated. That prevents most chromosome-counting traps.

Question	A cell with $2n = 10$ undergoes meiosis. Nondisjunction affects one chromosome pair. (a) Predict the chromosome number in each gamete if nondisjunction occurs in meiosis I. (b) Predict the chromosome number in each gamete if nondisjunction occurs in meiosis II.
Step 1 — Identification	Determine the haploid number and identify whether the failed separation is homologous chromosomes in meiosis I or sister chromatids in meiosis II.
Step 2 — Principle	Meiosis I nondisjunction affects both daughter cells, so all four gametes become abnormal. Meiosis II nondisjunction affects only one of the two meiosis-II divisions, so two gametes stay normal and two become abnormal.
Step 3 — Explicit Reasoning	Because $2n = 10$, $n = 5$. (a) Meiosis I nondisjunction: homologous chromosomes fail to separate in the first division. One cell gets both homologs and one gets none. After meiosis II, this produces two gametes with $n + 1 = 6$ and two gametes with $n - 1 = 4$. (b) Meiosis II nondisjunction: meiosis I is normal, so both cells entering meiosis II have $n = 5$. In one of those cells, sister chromatids fail to separate. Final products are two normal gametes with 5 chromosomes, one gamete with 6, and one gamete with 4.
Step 4 — Validation	The pattern should reveal timing: an earlier error spreads to every product, while a later error affects only half the products.
Step 5 — Final Answer	Meiosis I nondisjunction: 6, 6, 4, 4. Meiosis II nondisjunction: 5, 5, 6, 4.
Common Error	Saying meiosis II nondisjunction makes all four gametes abnormal.
Exam Strategy	Memorize the diagnostic pattern, then justify it: meiosis I error $\rightarrow$ 4 abnormal gametes; meiosis II error $\rightarrow$ 2 normal and 2 abnormal.

Question	In a testis sample, 75 primary spermatocytes enter meiosis. Assuming 100% survival and normal meiosis, determine: (a) the number of secondary spermatocytes formed (b) the number of spermatids formed (c) the number of sperm cells produced
Step 1 — Identification	Track one primary spermatocyte through both meiotic divisions and then through maturation.
Step 2 — Principle	Each primary spermatocyte completes meiosis I to form two secondary spermatocytes. Each secondary spermatocyte completes meiosis II, so one primary spermatocyte yields four spermatids, which mature into four sperm.
Step 3 — Explicit Reasoning	Secondary spermatocytes = $75 \times 2 = 150$.

	Spermatids = $75 \times 4 = 300$. Because maturation is 1:1, sperm cells = 300.
Step 4 — Validation	The multiplication should follow the biological sequence 1 → 2 → 4, not an unexplained arithmetic shortcut.
Step 5 — Final Answer	(a) 150 secondary spermatocytes (b) 300 spermatids (c) 300 sperm cells
Common Error	Multiplying by two twice without stating which meiotic event causes each increase.
Exam Strategy	Write the chain first: 1 primary → 2 secondary → 4 spermatids → 4 sperm. Then scale by the number of starting cells.

Question	A human female has 12 primary oocytes completing meiosis (ignore arrests and assume completion). Determine: (a) the maximum number of functional ova produced (b) the maximum number of polar bodies produced
Step 1 — Identification	Identify that this question is about oogenesis rather than spermatogenesis, so product number depends on unequal cytokinesis.
Step 2 — Principle	Oogenesis conserves almost all cytoplasm in one cell to support early embryonic development. Therefore, one primary oocyte yields one functional ovum, while the remaining meiotic products become polar bodies.
Step 3 — Explicit Reasoning	Functional ova = 12 primary oocytes × 1 ovum per primary oocyte = 12 ova. Using the standard diploma model, one primary oocyte can produce three polar bodies in total. Polar bodies = $12 \times 3 = 36$.
Step 4 — Validation	The chromosome mechanism of meiosis is still the same as in males, but the cytoplasmic outcome is different because reproductive function is different.
Step 5 — Final Answer	(a) 12 functional ova (b) 36 polar bodies
Common Error	Answering four ova because meiosis produces four cells structurally.
Exam Strategy	Separate chromosome behaviour from cytoplasmic distribution. Meiosis can make four cells structurally, but only one becomes the functional ovum.

Question	A tetrad undergoes crossing over between non-sister chromatids in prophase I. One crossover event occurs. (a) How many chromatids in the tetrad become recombinant? (b) Explain why the other chromatids may remain non-recombinant.
Step 1 — Identification	Define the structure involved. A tetrad contains two homologous chromosomes, and each homolog has two sister chromatids.
Step 2 — Principle	A single crossover usually occurs between two non-sister chromatids. Only the chromatids that physically exchange DNA segments become recombinant.

Step 3 — Explicit Reasoning	A tetrad contains four chromatids total. If one crossover occurs between one chromatid from each homolog, exactly two chromatids exchange corresponding DNA segments. Those two chromatids become recombinant. The other two chromatids did not participate in the exchange, so they retain their original allele combinations and remain parental.
Step 4 — Validation	Crossing over changes sequence arrangement on participating chromatids, but it does not change chromosome number or ploidy.
Step 5 — Final Answer	(a) Two chromatids become recombinant. (b) The other two chromatids remain non-recombinant because they did not participate in the DNA exchange.
Common Error	Saying all four chromatids become recombinant after one crossover.
Exam Strategy	Count the chromatids that actually touch and exchange segments. One crossover event usually alters only two chromatids.

Question	An onion root tip sample contains 600 cells. Observations show the following stage counts: Interphase 450, Prophase 60, Metaphase 45, Anaphase 30, Telophase 15. Assuming a 24-hour cell cycle, estimate the time spent in each stage.
Step 1 — Identification	Recognize this as a data-analysis question. The snapshot count in each stage is used to estimate the fraction of the full cell cycle spent in that stage.
Step 2 — Principle	If cells move through the cycle continuously, the proportion of cells observed in a stage is approximately equal to the proportion of total time spent in that stage.
Step 3 — Explicit Reasoning	Total cells = 600 and total cycle time = 24 h. Interphase: $450/600 = 0.75 \rightarrow 0.75 \times 24 = 18.0$ h Prophase: $60/600 = 0.10 \rightarrow 0.10 \times 24 = 2.4$ h Metaphase: $45/600 = 0.075 \rightarrow 0.075 \times 24 = 1.8$ h Anaphase: $30/600 = 0.05 \rightarrow 0.05 \times 24 = 1.2$ h Telophase: $15/600 = 0.025 \rightarrow 0.025 \times 24 = 0.6$ h
Step 4 — Validation	Add the stage times to verify the estimate: $18.0 + 2.4 + 1.8 + 1.2 + 0.6 = 24.0$ h, so the calculations are internally consistent.
Step 5 — Final Answer	Estimated stage times: Interphase 18.0 h Prophase 2.4 h Metaphase 1.8 h Anaphase 1.2 h Telophase 0.6 h
Common Error	Multiplying by 24 before converting each count to a fraction of the total number of cells.
Exam Strategy	For stage-count questions, always do fraction first, time second, and then perform a sum check.

Question	A student claims, "Meiosis is just mitosis done twice, so the products are basically the same." Evaluate this claim using at least four specific biological differences.
Step 1 — Identification	Identify that the task is evaluative. A correct answer must compare mechanism and outcome, not just list that meiosis has two divisions.
Step 2 — Principle	Processes are not equivalent when they differ in chromosome pairing, DNA exchange, separation pattern, ploidy outcome, and genetic consequence.
Step 3 — Explicit Reasoning	Meiosis includes synapsis of homologous chromosomes in prophase I; mitosis does not. Meiosis includes crossing over between non-sister chromatids; mitosis typically does not. Meiosis I separates homologous chromosomes, causing reduction from $2n$ to n; mitosis separates sister chromatids and maintains chromosome number. Meiosis produces four genetically varied haploid cells; mitosis produces two genetically similar diploid cells used for growth and repair. Therefore, meiosis is not simply mitosis repeated twice; it is a specialized reduction-and-recombination process.
Step 4 — Validation	A valid evaluation must connect each mechanistic difference to its effect on chromosome number or genetic variation.
Step 5 — Final Answer	The claim is incorrect. Meiosis differs from mitosis in homologous pairing, crossing over, reduction division, chromosome-number outcome, and the number and genetic diversity of final products.
Common Error	Listing only 'two divisions versus one division' without explaining why the outcomes differ.
Exam Strategy	Use a compare-and-conclude pattern: 'Meiosis has ____; mitosis does not; therefore the products differ because ____.

Question	Explain how monozygotic (identical) twins can have different phenotypes even though they have the same genotype. Provide two biologically valid mechanisms.
Step 1 — Identification	Distinguish genotype from phenotype. Genotype refers to DNA sequence, while phenotype depends on gene expression and developmental conditions.
Step 2 — Principle	Phenotype can change without changing inherited alleles if environmental conditions alter development or if gene expression changes through epigenetic regulation. Somatic mutation after zygote splitting can also create divergence.
Step 3 — Explicit Reasoning	Mechanism 1: Environmental differences. Different placental blood supply, nutrient access, or exposure to teratogens can alter developmental pathways and produce different traits. Mechanism 2: Epigenetic differences. DNA methylation or histone modification can change which genes are expressed, even when both twins carry the same alleles. A third valid mechanism is somatic mutation after the embryo splits, which can create genetic mosaicism in one twin.
Step 4 — Validation	The reasoning remains consistent because none of these mechanisms requires different parental gametes. The twins can begin with the same zygotic genotype and still diverge later.

Step 5 — Final Answer	Monozygotic twins can differ phenotypically because development is influenced by environment and epigenetic regulation, and in some cases by post-zygotic somatic mutation.
Common Error	Assuming identical genotype guarantees identical phenotype in every circumstance.
Exam Strategy	When phenotype differs but genotype is shared, think environment, gene expression, and mutation timing.

Question	Researchers compare two plant life cycles using the observations below. Organism A: the large visible stage is haploid; a smaller diploid capsule remains attached and releases spores. Organism B: the large visible stage is diploid; microscopic haploid pollen and female gametophyte structures depend on the larger plant. Interpret the data by identifying which organism is moss and which is pine, stating the dominant stage in each, and explaining what 'dominant' means in this context.
Step 1 — Identification	This is a life-cycle data-interpretation question. Use the observed ploidy and visibility of each stage to identify gametophyte-dominant versus sporophyte-dominant organisms.
Step 2 — Principle	In alternation of generations, 'dominant' means the stage that is larger, longer-lived, and most independent. Moss has a dominant gametophyte, whereas pine has a dominant sporophyte.
Step 3 — Explicit Reasoning	Organism A has a large visible haploid stage and a smaller attached diploid structure that releases spores. That pattern matches moss, where the gametophyte is the dominant generation and the sporophyte is dependent. Organism B has a large visible diploid stage and highly reduced haploid structures such as pollen and the female gametophyte. That pattern matches pine, where the sporophyte is dominant and the gametophytes are reduced and dependent.
Step 4 — Validation	Check the ploidy logic: sporophytes are diploid and make spores by meiosis; gametophytes are haploid and make gametes by mitosis.
Step 5 — Final Answer	Organism A is moss, with a dominant haploid gametophyte. Organism B is pine, with a dominant diploid sporophyte. Here, 'dominant' means the larger, longer-lived, and more independent stage of the life cycle.
Common Error	Mixing up spores and gametes or assuming 'dominant' means genetically dominant.
Exam Strategy	Translate the observation into ploidy first, then identify whether the prominent stage is gametophyte (n) or sporophyte (2n).

Question	A targeted cancer therapy is designed to stop rapidly dividing cells by preventing spindle fibre formation. (a) Identify the mitotic phase most directly affected. (b) Predict what would be observed in treated cells. (c) Explain one reason why this therapy can also harm healthy tissues.
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Step 1 — Identification	Identify the structure being targeted. Spindle fibres are needed for chromosome attachment, alignment, and separation.
Step 2 — Principle	A process fails most obviously at the stage where the blocked structure is required to complete the next transition. Without spindle fibres, chromosomes cannot align and segregate properly, so the spindle checkpoint stops progression.
Step 3 — Explicit Reasoning	(a) The phase most directly affected is metaphase, especially the transition into anaphase, because chromosomes cannot achieve proper bipolar attachment and alignment. (b) Treated cells would accumulate in mitotic arrest with condensed chromosomes that are not properly aligned or segregated. Many cells would later undergo apoptosis. (c) Healthy tissues such as hair follicles, intestinal lining, and bone marrow also divide rapidly, so they are harmed because the drug targets cell-division machinery rather than cancer cells alone.
Step 4 — Validation	This answer fits checkpoint logic: if spindle attachment fails, the cell should arrest before successful anaphase rather than completing normal chromosome separation.
Step 5 — Final Answer	(a) Metaphase, especially the metaphase-to-anaphase transition (b) Mitotic arrest with condensed chromosomes and failed segregation (c) Damage to normal rapidly dividing tissues because they also require spindle fibres for mitosis
Common Error	Answering only 'prophase' because spindle formation begins there.
Exam Strategy	Ask which stage requires the targeted structure to complete successfully. Then predict arrest at that checkpoint-dependent stage.